

SUNDSVALL TIMRA, Sweden

WMO: 023660

Lat: 62.528N

Lon: 17.444E

Elev: 5

StdP: 101.27

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.2	-23.9	0.4	-21.6	-21.2	0.6	-19.1	11.5	0.1	10.0	-1.3	1.5	310	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	24.8	17.5	22.9	16.6	21.2	15.7	18.8	22.7	17.7	21.2	16.7	19.9	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.4	20.1	16.2	11.5	19.0	15.2	10.8	18.0	53.3	22.7	49.9	21.2	46.9	20.0	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	7.0	DB	-25.0	27.8	3.8	2.3	-27.7	29.5	-29.9	30.8	-32.0	32.1	-34.8	33.8	(4)
(5)			WB	-25.2	19.8	3.7	1.2	-27.9	20.7	-30.0	21.4	-32.1	22.1	-34.8	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-6.6	-6.2	-2.0	2.8	7.9	12.9	16.2	14.9	10.5	4.5	-0.7	-4.6	(6)	
	DBStd	8.88	5.93	6.12	4.52	2.95	3.30	2.72	2.58	2.66	2.91	3.20	4.02	5.51	(7)	
	HDD10.0	2616	513	453	371	216	81	6	0	1	29	172	322	452	(8)	
	HDD18.3	5177	772	686	629	466	322	164	78	112	236	429	572	711	(9)	
	CDD10.0	497	0	0	0	0	17	92	191	152	43	2	0	0	(10)	
	CDD18.3	16	0	0	0	0	0	1	11	4	0	0	0	0	(11)	
	CDH23.3	125	0	0	0	0	2	13	81	28	1	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	1	10	2	0	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.5	2.5	2.9	2.9	3.2	3.4	3.1	2.9	2.8	2.7	2.6	2.5	(14)	
(15) Precipitation	PrecAvg	668	53	33	33	38	39	57	78	84	65	69	62	57	(15)	
	PrecMax	1003	126	67	75	109	91	123	190	160	202	183	203	109	(16)	
	PrecMin	500	9	5	8	9	5	18	10	8	15	13	12	4	(17)	
	PrecStd	120	28	16	17	26	20	27	45	40	42	44	43	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	8.1	11.0	16.9	22.0	25.1	27.9	26.1	21.2	14.4	10.2	7.1	(19)	
		MCWB	3.7	4.5	6.1	9.6	13.7	16.4	18.4	17.9	15.3	10.8	7.7	4.9	(20)	
	2%	DB	3.9	5.9	8.4	13.8	18.9	22.4	25.5	23.8	18.8	12.2	7.5	5.1	(21)	
		MCWB	2.0	3.3	4.3	7.9	11.9	14.7	17.8	17.5	14.4	9.8	6.5	3.4	(22)	
	5%	DB	2.2	3.6	6.6	11.0	16.8	20.7	23.7	21.9	17.1	11.1	6.1	3.7	(23)	
		MCWB	1.2	1.5	3.2	6.1	10.9	14.0	17.3	16.7	13.4	9.3	5.4	2.7	(24)	
	10%	DB	1.1	1.6	4.7	9.0	14.8	18.9	22.0	20.2	15.8	9.9	5.0	2.2	(25)	
		MCWB	0.5	0.5	1.8	5.0	9.4	13.1	16.3	15.7	12.8	8.2	4.2	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.8	6.6	9.9	14.7	17.5	20.4	20.2	16.7	12.2	8.3	5.2	(27)	
		MCDB	5.8	7.1	10.3	15.6	20.2	22.0	24.1	23.7	19.3	13.4	9.5	6.3	(28)	
	2%	WB	2.4	3.3	4.7	8.2	12.9	16.0	19.2	18.7	15.4	10.9	6.6	3.8	(29)	
		MCDB	3.3	5.6	7.9	12.9	17.7	20.7	23.5	21.7	17.5	11.7	7.1	4.6	(30)	
	5%	WB	1.4	1.7	3.3	6.6	11.3	14.9	18.2	17.6	14.3	9.6	5.5	2.6	(31)	
		MCDB	2.1	2.9	6.1	10.4	15.7	18.8	22.2	20.3	16.2	10.7	5.9	3.4	(32)	
	10%	WB	0.5	0.6	2.1	5.2	10.0	13.9	17.2	16.6	13.4	8.3	4.3	1.5	(33)	
		MCDB	1.0	1.4	4.2	8.2	13.7	17.6	20.7	19.3	15.2	9.4	4.8	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.7	8.9	9.4	10.0	9.4	9.1	9.0	8.6	7.2	5.4	6.4	(35)	
		MCDBR	8.4	10.1	11.1	14.1	13.8	12.8	12.0	11.3	10.5	9.4	6.7	7.7	(36)	
	5% WB	MCWBR	7.3	7.8	7.5	8.5	7.4	6.3	5.7	5.8	6.5	6.8	5.7	6.5	(37)	
		MCWBR	7.9	8.9	10.5	13.2	12.7	10.9	10.4	9.6	9.2	8.1	6.0	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.240	0.297	0.320	0.358	0.359	0.354	0.371	0.363	0.329	0.315	0.258	0.346	(40)	
		taud	2.103	2.235	2.333	2.360	2.424	2.473	2.445	2.463	2.541	2.436	2.175	2.595	(41)	
	Edn at Noon		424	650	786	827	859	871	842	817	784	640	396	256	(42)	
		Edh at Noon	32	62	82	98	100	97	98	88	68	52	30	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.20	0.84	2.31	3.76	4.79	5.54	5.06	3.84	2.41	1.05	0.30	0.11	(44)	
	RadStd		0.03	0.14	0.26	0.32	0.49	0.35	0.55	0.32	0.30	0.19	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.52	N/A	N/A	N/A	N/A	N/A	-149	-166	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page